

Research Methodology in CS (CSEG3060_3)

Lecture 3: Patenting Process - Research and Innovation: Stages of prior-art search, novelty assessment, invention disclosure, drafting of patent specification, and role of research and innovation in the patenting process.

Unit 5: Intellectual Property and Patents • Mapped Course Outcome: CO2

Lecture 3: Patenting Process — Research and Innovation

Course: Research Methodology in CS (CSEG3060_3) **Program:** B.Tech (Computer Science) **Semester:** 5 **Unit:** 5 — Intellectual Property and Patents
Mapped Course Outcome: CO2 **Lecture Duration:** 60 minutes (recommended)

1. Lecture Learning Outcomes

By the end of this lecture, the student should be able to:

1. **Describe** the sequential and iterative workflow that transforms a research idea into a filed patent application.
2. **Conduct** a structured prior-art search using appropriate patent and non-patent databases.
3. **Apply** statutory criteria of novelty to assess whether an invention qualifies for patent protection.
4. **Prepare** a competent invention disclosure that captures all the technical information required by the patent attorney.
5. **Draft** the structural and linguistic elements of a complete patent specification, including independent and dependent claims.
6. **Evaluate** the strategic role of research and innovation in building a defensible patent portfolio.

2. Introduction and Pedagogical Context

The previous lectures in this unit established the institutional and conceptual scaffolding of intellectual property (IP) — the categories of IP, the philosophy of statutory protection, and the international framework administered under the World Intellectual Property Organization (WIPO) and the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS). Lecture 3 moves from the *why* of IP to the *how* of patenting.

For a computer science researcher, the patenting process is rarely a downstream, post-thesis formality. In contemporary practice, especially in domains such as artificial intelligence, cryptography, networking protocols, and hardware-software co-design, patenting decisions are made *during* the research lifecycle. Researchers publish, file patent applications, and protect trade secrets in a parallel and tightly coupled workflow. Understanding this workflow is therefore a core research methodology skill, not a peripheral legalistic exercise.

This lecture traces the patenting pipeline from the moment an inventor believes they have solved a technical problem to the moment a complete specification is filed with a patent office. The pipeline has four operational pillars:

- **Prior-art search**
- **Novelty assessment**
- **Invention disclosure**
- **Drafting of the patent specification**

Each pillar is examined in depth, with a closing section on how research and innovation management frame the entire activity.

3. Key Concepts and Terminology

The following terms will be used throughout the lecture. They are introduced here for clarity.

Term	Definition
Prior Art	Any public disclosure, anywhere in the world, before the filing date (or priority date) that may anticipate the claimed invention.
Novelty	The statutory requirement that the invention must not have been disclosed in any prior art.
Inventive Step / Non-Obviousness	The requirement that the invention, as claimed, would not be obvious to a person skilled in the art (PSA).
Invention Disclosure	A confidential, internal document prepared by the inventor describing the invention in sufficient technical detail to enable patent prosecution.
Patent Specification	The legally binding document filed with the patent office, comprising the description, claims, abstract, and drawings.
Provisional Specification	A placeholder application that establishes the priority date but does not require claims.
Complete Specification	The full patent document filed within 12 months of the provisional, including claims.
Priority Date	The filing date of the earliest application on the basis of which subsequent applications claim priority.
PCT	The Patent Cooperation Treaty, administered by WIPO, providing a unified procedure for international filing.
Patentability	The umbrella term for meeting the statutory requirements of novelty, inventive step, and industrial application.
PSA (Person Skilled in the Art)	A hypothetical ordinary practitioner in the relevant technical field used to judge obviousness.
FTO (Freedom to Operate)	An analysis confirming that a product or process does not infringe the claims of existing patents.

4. Detailed Explanations

4.1 Stage 1: Prior-Art Search

4.1.1 Definition and Legal Significance

A prior-art search is a systematic retrieval and analysis of existing public disclosures that are relevant to a proposed invention. Section 2(1)(l) of the Indian Patents Act, 1970 (and corresponding statutes worldwide) defines "prior art" as:

"any matter which has been published or disclosed anywhere in the world, before the date of filing of the patent application, including prior claiming."

The legal significance is twofold. First, prior art is used by patent examiners to assess novelty and inventive step. Second, it is used by applicants and their attorneys to evaluate the probability of grant before filing, thereby avoiding wasted expenditure.

4.1.2 Types of Prior Art

- **Patent literature:** Granted patents and published patent applications filed with national and regional patent offices.
- **Non-patent literature (NPL):** Peer-reviewed journals, conference proceedings (e.g., ACM, IEEE), theses, textbooks, technical reports, standards documents (e.g., RFCs), and product manuals.
- **Public disclosures:** Conference talks, posters, blog posts, open-source code commits, YouTube demonstrations, and marketing material.
- **Public use or sale:** In many jurisdictions, including the United States under 35 U.S.C. § 102(a)(1), prior public use or sale constitutes prior art regardless of whether it was documented in writing.

In the CS context, open-source software repositories on GitHub, preprints on arXiv, and standardisation documents from bodies such as the IETF, IEEE, and 3GPP are increasingly important sources of prior art.

4.1.3 Search Tools and Databases

Database	Coverage	Strength
Google Patents	Global, including WIPO data	Free, full-text search, citation graph
USPTO PAIR / Patent Full-Text and Image Database (PatFT/AppFT)	United States	Authoritative US source
Espacenet (EPO)	Global, 100+ countries	Multilingual, classification search
WIPO PATENTSCOPE	PCT applications	Authoritative PCT source
IEEE Xplore	CS/EE conference and journal literature	Best for CS non-patent literature
ACM Digital Library	CS conference proceedings and journals	Best for software-related prior art
ScienceDirect	Multidisciplinary scientific literature	Cross-domain coverage
Indian Patent Database (IP India)	Indian filings and grants	Domestic filing and opposition

4.1.4 Search Strategy

A defensible prior-art search combines three orthogonal strategies:

1. **Keyword-based search:** Uses Boolean operators (AND, OR, NOT), proximity operators, and truncation. Example for a novel caching scheme:

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("cache replacement" OR "eviction policy") AND
("machine learning" OR "neural network") AND
("distributed" OR "edge computing")
```

2. **Classification-based search:** Uses Cooperative Patent Classification (CPC) or International Patent Classification (IPC) codes. For example, caching in distributed systems may fall under G06F12/0866 (cache hierarchies) or H04L67/568 (caching at network nodes).

3. **Citation-based search:** Forward and backward citation chasing. Once a relevant patent is identified, the cited references and citing patents are inspected iteratively.

The most rigorous searches combine all three strategies with iterative refinement. Search logs must be retained, since they form part of the due-diligence record required by the duty of candor before the USPTO and the disclosure obligations under Article 5 of the PCT.

4.1.5 Outcome of Prior-Art Search

The search produces a **prior-art landscape report** containing:

- Closest prior art (the most relevant documents).
- Categorisation by relevance (X, Y, A designations in EPO/IN search opinions).
- Identified gaps that the invention fills.
- Identified barriers if the invention lies within a heavily patented space.

This report feeds directly into the next stage: novelty assessment.

4.2 Stage 2: Novelty Assessment

4.2.1 The Statutory Standard

Under Section 2(1)(j) of the Indian Patents Act, 1970:

"invention" means a new product or process involving an inventive step and capable of industrial application.

"New" in this definition is interpreted as **absolute or global novelty** — the invention must not form part of the prior art anywhere in the world, in any language, and in any form, before the filing or priority date.

This contrasts with jurisdictions such as the United States, which historically applied a "first-to-invent" rule (modified by the America Invents Act, 2011, which aligned the US with the global "first-to-file" standard).

4.2.2 The Novelty Test

The novelty test is **claim-by-claim**. For each claim, the assessor asks:

Does a single prior-art reference disclose every element of the claim, either expressly or implicitly?

This is sometimes called the "**single-reference anticipation**" rule. If a single document contains all the elements of the claim, anticipation is established and the claim lacks novelty, even if the document does not explicitly state that those elements function together as claimed.

For software and CS inventions, novelty assessment must be careful about abstract ideas. In *Alice Corp. v. CLS Bank International*, 573 U.S. 208 (2014), the US Supreme Court established a two-step test (the "Alice/Mayo framework") to determine whether a software claim is directed to a patent-eligible abstract idea. Pure algorithms, mathematical formulas, and abstract business methods, even if novel, are not patent-eligible subject matter under 35 U.S.C. § 101. This restricts the patentability of certain classes of CS inventions and must be considered alongside novelty assessment.

4.2.3 Novelty vs. Inventive Step (Non-Obviousness)

A common confusion arises between novelty and inventive step. The distinction is critical:

- **Novelty** is binary: a claim is either anticipated or not.
- **Inventive step** is a sliding scale assessed against the PSA. Under Section 2(1)(ja) of the Indian Patents Act, an inventive step is a feature that:

"would not be obvious to a person skilled in the art."

Formally, given the prior art $\mathcal{P}$ and the invention I , novelty requires $I \not\subseteq \mathcal{P}$, while inventive step requires that I would not be reached by combining elements of $\mathcal{P}$ using ordinary creativity. In set-theoretic notation:

$I \text{ is novel} \iff \nexists p \in \mathcal{P} \supseteq I \text{ is novel} \iff \nexists p \in \mathcal{P} : p \supseteq I$

$I \text{ involves an inventive step} \iff I \text{ is not an obvious combination of elements in } \mathcal{P}$

$I \text{ involves an inventive step} \iff I \text{ is not an obvious combination of elements in } \mathcal{P}$

The two tests are complementary, and patentability requires both.

4.2.4 Assessment Procedure

A rigorous novelty assessment follows this protocol:

1. Construct the broadest reasonable interpretation (BRI) of each claim.
2. Identify the elements of each claim.
3. Map each element to specific passages in the closest prior art.
4. Document element-by-element comparison.
5. Flag any claim where a single reference anticipates all elements.
6. Flag any claim where anticipation can only be shown by combining references (this moves the analysis to inventive step).
7. Record findings in an invention review report.

If the assessment shows no anticipation and reasonable grounds for inventive step, the invention proceeds to disclosure and drafting. If anticipation is identified, the inventors may revise the claims to introduce differentiating features — a process known as **claim sculpting** — before proceeding.

4.3 Stage 3: Invention Disclosure

4.3.1 Definition and Purpose

An **invention disclosure** is a confidential written record prepared by the inventor(s) describing the invention, its technical context, the problem solved, and the contributors. It is the bridge between research output and legal filing. Its purposes are:

- To capture inventive details while still fresh in the inventor's mind.
- To establish internal priority among co-inventors and across competing internal projects.
- To provide the patent attorney with sufficient technical disclosure to draft a defensible specification.

- To comply with institutional IP policies of universities, R&D centres, and corporations.
- To serve as evidence in disputes over inventorship, ownership, or priority.

4.3.2 Timing of Disclosure

In an academic setting, the inventor must file the disclosure **before any external public disclosure** (including paper submission, conference presentation, or thesis deposit) that might compromise novelty. Most universities have technology transfer offices (TTOs) that review disclosures confidentially and recommend a publication embargo if a patent filing is anticipated.

In an industry setting, the disclosure is filed through an internal invention management system, often within 30 days of conception, as required by many employer invention assignment policies.

4.3.3 Components of an Invention Disclosure Form

A typical IDF (Invention Disclosure Form) used in academic and industry TTOs contains the following sections:

1. **Title of the Invention:** Concise, descriptive, non-marketing.
2. **Inventor(s):** Full legal names, affiliations, contribution percentages. Note: inventorship is a legal determination; the IDF lists candidate inventors.
3. **Funding Sources:** Grants, contracts, sponsors — these affect ownership and government rights (e.g., under the Bayar Act in the US or Section 7 of the Indian Patents Act).
4. **Problem Solved:** The technical problem addressed by the invention.
5. **Prior Art Known to the Inventors:** Any references the inventors are aware of that are relevant to novelty.
6. **Detailed Technical Description:** Enough detail for a person skilled in the art to reproduce the invention — the "enablement" standard.
7. **Best Mode:** The inventors' preferred embodiment at the time of filing.
8. **Drawings, Flowcharts, Block Diagrams:** Visual supplements to the written description.
9. **Keywords and Suggested CPC/IPC Classifications:** To guide prior-art searching.
10. **Commercial Application:** Anticipated industries, products, and licensees.
11. **Public Disclosures to Date:** Including dates of any talks, papers, posters, code releases.
12. **Confidentiality Statement:** Signed by all inventors acknowledging confidentiality.

4.3.4 Common Pitfalls in Invention Disclosure

- **Late disclosure** after public presentation, which destroys novelty in absolute-novelty jurisdictions.
- **Insufficient detail** that prevents the attorney from drafting a defensible specification.
- **Improper inventorship listing**, e.g., omitting a contributor who contributed to the conception of a claim element, or including a supervisor who did not contribute.
- **Failure to identify funding sources**, which may create government ownership rights (notably under the Bayar-Dole Act for US-funded research).
- **Premature external disclosure** during commercialisation discussions without a non-disclosure agreement (NDA).

4.4 Stage 4: Drafting the Patent Specification

4.4.1 Provisional vs. Complete Specification

The Indian Patents Act (and analogous legislation elsewhere) permits a two-stage filing:

- **Provisional Specification** (Section 9): Establishes the priority date and contains a description of the invention. No claims are required. Used when the invention is at an early stage and the applicant needs additional time to refine.
- **Complete Specification** (Section 10): Filed within 12 months of the provisional. Contains the full description, abstract, and claims. The complete specification is the document on which examination and grant are based.

The 12-month window between provisional and complete filings is a strategic period. The applicant gains priority for what is disclosed in the provisional, but can refine the claims in the complete specification to encompass improvements made during the priority year.

4.4.2 Structure of a Complete Specification

The complete specification follows the structure prescribed by Rule 13 of the Indian Patent Rules, 2003 (with parallels in 37 CFR § 1.77 in the US and Rule 42 EPC in Europe):

1. **Title of the Invention** — concise, technically descriptive, free of trade names.

2. **Field of the Invention** — broad classification of the technical domain.
3. **Background of the Invention** — description of the prior art and the problem addressed.
4. **Objects of the Invention** — explicit statements of what the invention seeks to achieve.
5. **Summary of the Invention** — concise overview of the solution.
6. **Brief Description of the Drawings** — brief explanation of each figure.
7. **Detailed Description of the Invention** — full technical description, including embodiments and best mode.
8. **Claims** — the legally operative section defining the scope of protection.
9. **Abstract** — a 150-word summary used for prior-art retrieval.

4.4.3 Drafting the Claims

Claims are the most critical part of the specification. They define, with legal precision, the scope of the monopoly. There are two principal types:

- **Independent claims:** Stand alone, define the broadest inventive concept.
- **Dependent claims:** Reference one or more preceding claims, add further limitations, and provide fallback positions if the independent claim is invalidated.

In CS-related patents, claims may be categorised as:

- **Method claims:** Define a sequence of steps (e.g., a method for training a neural network).
- **System claims:** Define an apparatus comprising certain components (e.g., a distributed system for federated learning).
- **Computer-readable medium claims:** Define a non-transitory medium storing instructions (e.g., a non-transitory computer-readable storage medium storing instructions that, when executed, perform a process).
- **Use claims:** Define a new use of a known device or composition (less common in CS, more common in pharmaceuticals).

A well-drafted claim follows a consistent structural template:

"A [method/system/computer-readable medium] for [purpose], comprising: (a) [first element]; (b) [second element]; and (c) [third element]."

Two drafting principles govern claim language:

- **Definiteness (35 U.S.C. § 112 / Indian Section 4(4)(iii)):** Claims must particularly point out and distinctly claim the subject matter. Functional language without structural support is generally rejected as indefinite.
- **Support (Section 10(4)):** Claims must be supported by the description. Claim features absent from the description are not allowable.

4.4.4 Best Practices in Specification Drafting

- Use consistent terminology throughout the description and claims.
- Avoid relative terms such as "large" or "fast" without quantitative anchors.
- Define key terms explicitly in the description (definition section).
- Disclose alternative embodiments even if not currently claimed — they provide fallback positions and support for dependent claims.
- Provide working examples sufficient to satisfy the enablement requirement.
- Coordinate claim scope with commercial value: too broad invites rejection; too narrow is easily designed around.
- Maintain a direct correspondence between claim features and description paragraphs.

4.4.5 Common Drafting Errors

- **Anticipation by the description:** The specification itself discloses the invention in a manner that anticipates the claims.
- **New matter:** Introducing matter into the complete specification not present in the provisional — this destroys priority.
- **Improper dependent claim formatting:** A dependent claim that does not further limit the parent claim.
- **Multiple inventions (lack of unity):** A single application claiming disparate inventions, triggering a unity-of-invention objection under PCT Rule 13.

4.5 Role of Research and Innovation in the Patenting Process

4.5.1 Research as the Source of Invention

The patenting process is fundamentally driven by the research pipeline. The flow can be modelled as:

$$R \xrightarrow{\text{conception}} I \xrightarrow{\text{disclosure}} D \xrightarrow{\text{drafting}} S \xrightarrow{\text{examination}} G$$

R conception

I disclosure

D drafting

S examination

G

where *RR* is the research output, *II* is the invention, *DD* is the disclosure, *SS* is the specification, and *GG* is the granted patent. Each transition requires rigorous methodological discipline.

Patentable inventions typically emerge from research of Technology Readiness Level (TRL) 3–7 — that is, from proof-of-concept through to prototype demonstration. Lower-TRL research (basic science) and higher-TRL work (manufacturing scale-up) yield fewer patentable inventions, though they may produce other forms of IP (trade secrets, know-how, regulatory data).

4.5.2 Patent Landscape Analysis

Before committing resources to a research direction, sophisticated R&D organisations conduct a **patent landscape analysis** — a structured study of existing filings in the relevant technology area. Landscape analyses reveal:

- White spaces (underexplored technology areas).
- Crowded spaces (over-patented areas with high freedom-to-operate risk).
- Key assignees (competitors and collaborators).
- Citation clusters (technology hotspots).
- Filing trends (emerging vs. declining areas).

This analysis, increasingly performed with machine-learning tools, allows research to be directed toward patentable and strategically valuable problems.

4.5.3 Freedom to Operate (FTO)

Distinct from patentability (the ability to obtain a patent), **Freedom to Operate** is the ability to commercialise a product without infringing existing third-party patents. An FTO analysis identifies blocking patents whose claims read on the proposed product or process. CS researchers developing algorithms, software libraries, or protocols must conduct FTO analyses, often more stringently than patentability searches, because many jurisdictions (notably the US) restrict patent-eligibility of pure algorithms under § 101, but allow patents on specific applications of algorithms.

4.5.4 The Innovation Pipeline and Patent Quality

Not every research output warrants a patent. The decision matrix typically considers:

Dimension	Question
Technical Merit	Does the invention solve a real technical problem?
Novelty	Does it clear prior art?
Commercial Value	Is there a market, a licensee, or a strategic motive?
Defensibility	Can the claims withstand invalidation challenges?
Cost-Benefit	Are filing, prosecution, and maintenance costs justified?
Disclosure Risk	Is the technical information better kept as a trade secret?

In mature IP management frameworks (e.g., those aligned with ISO 56005 for innovation management), patent quality is preferred to patent quantity. A small portfolio of well-drafted, well-prosecuted, and well-maintained patents typically outperforms a large portfolio of marginal filings.

4.5.5 Research-Patent Synergy

In academic settings, the Bayar-Dole Act (US) and analogous Indian provisions (e.g., the Indian Council of Scientific and Industrial Research framework, the Public Funded Research Result Related IP Protection and Management Rules, 2023) allow universities to retain title to inventions arising from publicly funded research, subject to compliance with disclosure, election of title, and march-in rights. This has institutionalised the research-patent synergy, making patent awareness a core research methodology competency.

Computer science departments increasingly embed IP modules in graduate research training precisely because modern CS research — from federated learning to hardware security — produces a steady stream of patentable subject matter.

5. Practical Examples and Case Studies

Case Study A: Prior-Art Search in Action — Smartphone Touchscreen Patents

In the late 1990s, multiple inventors independently developed capacitive touchscreen technology. The prior-art landscape at the time of filings included FingerWorks (acquired by Apple in 2005), IBM patents on capacitive sensing, and academic publications in IEEE Transactions on Consumer Electronics. A rigorous prior-art search at the time of any new filing would have revealed the crowded landscape and shaped both claim scope and filing strategy. The litigation history of touchscreen patents — including the Apple v. Samsung disputes — illustrates the cost of inadequate landscape awareness.

Case Study B: Novelty and Software Patents — Alice Corp. v. CLS Bank

Alice Corporation held patents on a computerised method for mitigating settlement risk in financial transactions. The Supreme Court held that claims directed to the abstract idea of intermediated settlement, even when implemented on a generic computer, were not patent-eligible under § 101. The case reshaped CS patent strategy, requiring drafters to frame claims around specific technical improvements (e.g., "a method of training a neural network wherein the gradient update is computed using a novel pruning threshold") rather than abstract ideas implemented on a computer.

Case Study C: Indian CS Patent — BharatPi and Open-Source Disclosure

Several Indian startups developing single-board computers (e.g., BharatPi, an indigenous Raspberry Pi alternative) and similar CS hardware projects have had to balance open-source publication with patent strategy. Publishing schematics on GitHub before filing creates public disclosure that destroys novelty in jurisdictions requiring absolute novelty. This illustrates the importance of invention disclosure timing in the modern open-source era.

Case Study D: Drafting Quality — The "Page-Rank" Patent (Stanford v. Google)

The PageRank patent (US 6,285,999, "Method for node ranking in a linked database", assigned to Stanford, licensed exclusively to Google) demonstrates high-quality CS patent drafting. The claims are drafted around the specific mathematical formulation, the citation graph traversal, and the random surfer model — features that are defensible both against prior art and against § 101 abstract-idea challenges. The patent ultimately generated over \$300 million in licensing revenue for Stanford, illustrating the commercial value of research-driven patent drafting.

6. Summary

This lecture traced the patenting pipeline through its four operational pillars and examined the role of research and innovation as the upstream driver.

- **Prior-art search** is a structured retrieval across patent and non-patent literature using keyword, classification, and citation strategies. It informs both patentability and freedom-to-operate decisions.
- **Novelty assessment** applies the single-reference anticipation test to each claim, in light of absolute novelty requirements and subject-matter eligibility constraints such as the Alice/Mayo framework.
- **Invention disclosure** is the confidential bridge between research and legal filing, capturing inventorship, enablement, best mode, funding sources, and public disclosure history.
- **Drafting of the patent specification** requires careful coordination of provisional and complete filings, structured description, and carefully sculpted independent and dependent claims.
- **Research and innovation management** frames patenting decisions through landscape analysis, FTO, TRL alignment, and quality-over-quantity portfolio discipline.

The pipeline is iterative. Findings in one stage often force revision of an earlier stage — for example, prior-art search may require recasting claims, which in turn may require re-evaluating novelty and re-drafting the specification.

7. Exam Preparation Tips and Practice Questions

7.1 Frequently Tested Topics

- Definitions of prior art, novelty, inventive step, specification, claims.
- Distinction between novelty and inventive step.
- Sections of the Indian Patents Act governing novelty (Section 2(1)(j), 2(1)(l)) and specification (Sections 9, 10).
- Components of a complete specification.
- Types of claims.
- Role of person skilled in the art (PSA).
- Difference between patentability search and freedom-to-operate search.
- Alice/Mayo framework for software patent eligibility.

7.2 Short-Answer Questions (2–3 marks each)

1. Differentiate between novelty and inventive step with one example each from the computer science domain.
2. List five databases suitable for prior-art searching for software-related inventions.
3. What are the four main components of a complete patent specification? Briefly state the purpose of each.
4. State the difference between a provisional and a complete specification. Why is the provisional filing strategy important for CS researchers?
5. What is the "Person Skilled in the Art"? Why is this construct important in novelty assessment?
6. Explain why the *Alice Corp. v. CLS Bank* decision reshaped US software patent strategy.

7.3 Long-Answer / Analytical Questions (10–15 marks each)

1. A researcher at UPES develops a novel distributed caching protocol that uses federated reinforcement learning to adapt cache eviction policies to traffic patterns. Describe, stage by stage, the patenting pipeline the researcher should follow, identifying the deliverable of each stage and the potential pitfalls at each transition.
2. Compare the absolute novelty standard of the Indian Patents Act with the US first-inventor-to-file standard under the America Invents Act, 2011. Discuss the implications for software patent prosecution.
3. Critically examine the role of patent landscape analysis in computer science research. How does it influence problem selection, claim drafting, and commercialisation decisions?
4. "The claims of a patent define its legal scope; everything else is mere context." Critically evaluate this statement with reference to claim drafting principles, dependent claims, and the enablement requirement.
5. Draft an independent claim and three dependent claims for a hypothetical invention: *"A method for detecting adversarial perturbations in input images fed to a convolutional neural network, wherein the method computes a perturbation magnitude map and rejects inputs exceeding a dynamic threshold."*

7.4 Case-Based Practice

Read the following scenario and answer the questions below.

Dr. R. and two PhD students, A. and B., developed a novel routing protocol for low-earth-orbit satellite constellations. Before filing a patent, A. presented preliminary results at an international conference, with the slides uploaded to the conference website. Six months later, the team files a complete specification claiming the protocol. A year later, the patent examiner cites a 2019 paper by an unrelated group (X. et al.) that describes a similar routing principle but does not disclose the specific congestion-control mechanism of Dr. R.'s invention.

Questions:

(a) Identify any novelty issues arising from A.'s conference presentation. Under which jurisdictions is this fatal to patentability? (b) Can X. et al.'s 2019 paper anticipate Dr. R.'s claim? Apply the single-reference anticipation test. (c) Recommend corrective actions the team could have taken, and assess whether retroactive correction is possible.

7.5 Recommended Reading

- Indian Patents Act, 1970 (as amended) — Sections 2, 3, 4, 9, 10, 13, 25.

- WIPO. *Inventor Assistance Program* and *PCT Applicant's Guide* (latest edition).
 - World Intellectual Property Organization. *Drafting Patent Applications: A Practical Guide*.
 - Indian Patent Office Manual of Patent Practice and Procedure (latest edition).
 - *Alice Corp. v. CLS Bank International*, 573 U.S. 208 (2014).
 - *Diamond v. Chakrabarty*, 447 U.S. 303 (1980) — for biotechnology as an analogue to software subject matter eligibility.
 - Subramanian, S. *Patent Law: Cases and Materials*, LexisNexis (recent edition).
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8. Connection to Course Outcomes and Upcoming Lectures

This lecture directly advances **CO2**: the ability to apply research methodology principles to the protection and dissemination of research outputs. By tracing the patenting pipeline, students develop a procedural understanding that complements the substantive IP law covered in earlier lectures and the international cooperation mechanisms to be examined in the next lecture.

The next lecture will examine **international patenting cooperation**, focusing on the Patent Cooperation Treaty (PCT), regional offices (EPO, ARIPO, OAPI), and the procedural architecture of foreign filings. The drafting principles discussed today provide the foundation for that discussion, since international filings must harmonise with the requirements of multiple national offices simultaneously.